

RAVI SHANKAR(MATHS EXPERT)

TRIGONOMETRY

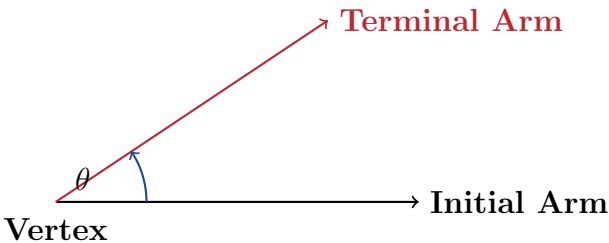
Class 11 – Complete Important Notes

Part 1: Angle & Radian Measure

1. What is an Angle?

CONCEPT

An **angle** is formed when a ray rotates about its initial point.
The fixed ray is called the **initial arm** and the rotated ray is called the **terminal arm**.
The point about which the rotation takes place is called the **vertex**.



2. Types of Angles

Type	Condition	Example
Zero angle	$\theta = 0^\circ$	0°
Acute angle	$0^\circ < \theta < 90^\circ$	30°
Right angle	$\theta = 90^\circ$	90°
Obtuse angle	$90^\circ < \theta < 180^\circ$	120°
Straight angle	$\theta = 180^\circ$	180°
Reflex angle	$180^\circ < \theta < 360^\circ$	270°
Complete angle	$\theta = 360^\circ$	360°

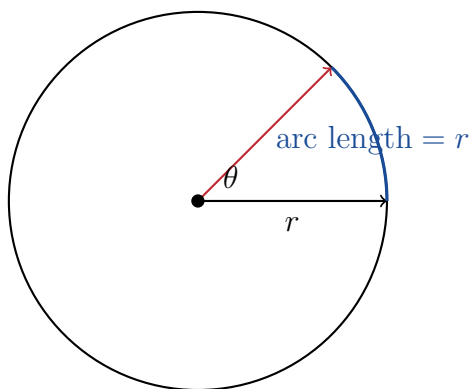
3. Degree Measure

CONCEPT

One complete revolution is divided into 360 equal parts.
Therefore,

$$1^\circ = \frac{1}{360}$$

of a complete revolution.
Thus,

**KEY FORMULA**

If

$$\text{Arc length} = r,$$

then the corresponding angle is

$$\boxed{1 \text{ radian}}.$$

5. Relation Between Degree and Radian

For a complete circle,

$$360^\circ = 2\pi \text{ radians}.$$

Therefore,

$$\boxed{180^\circ = \pi \text{ radians}}.$$

This is the most important conversion relation in Trigonometry.

KEY FORMULA

$$\boxed{180^\circ = \pi \text{ rad}}$$

Hence,

$$\boxed{1^\circ = \frac{\pi}{180} \text{ rad}}$$

and

$$\boxed{1 \text{ rad} = \frac{180^\circ}{\pi}}.$$

6. Degree to Radian Conversion

To convert degree into radian:

$$\boxed{\theta^\circ = \theta \frac{\pi}{180} \text{ rad}}$$

SOLVED EXAMPLE

Convert 60° into radians.

$$\begin{aligned}60^\circ &= 60 \times \frac{\pi}{180} \\&= \frac{\pi}{3}.\end{aligned}$$

Therefore,

$$60^\circ = \frac{\pi}{3} \text{ rad}$$

SOLVED EXAMPLE

Convert 225° into radians.

$$\begin{aligned}225^\circ &= 225 \times \frac{\pi}{180} \\&= \frac{5\pi}{4}.\end{aligned}$$

Therefore,

$$225^\circ = \frac{5\pi}{4} \text{ rad}$$

7. Radian to Degree Conversion

To convert radian into degree:

$$\theta \text{ rad} = \theta \times \frac{180^\circ}{\pi}$$

SOLVED EXAMPLE

Convert $\frac{5\pi}{6}$ radians into degrees.

$$\begin{aligned}\frac{5\pi}{6} \times \frac{180^\circ}{\pi} \\&= 5 \times 30^\circ \\&= \boxed{150^\circ}.\end{aligned}$$

SOLVED EXAMPLE

Convert $\frac{7\pi}{4}$ radians into degrees.

$$\begin{aligned}\frac{7\pi}{4} &\times \frac{180^\circ}{\pi} \\ &= 7 \times 45^\circ \\ &= \boxed{315^\circ}.\end{aligned}$$

8. Most Important Standard Conversions

Degree	Radian
0°	0
30°	$\frac{\pi}{6}$
45°	$\frac{\pi}{4}$
60°	$\frac{\pi}{3}$
90°	$\frac{\pi}{2}$
120°	$\frac{2\pi}{3}$
135°	$\frac{3\pi}{4}$
150°	$\frac{5\pi}{6}$
180°	π
270°	$\frac{3\pi}{2}$
360°	2π

9. Arc Length

If θ is measured in radians, then the length of the arc is

KEY FORMULA

$$\boxed{s = r\theta}$$

where

s = arc length, r = radius, θ = angle in radians.

IMPORTANT**Very Important:**

The formula

$$s = r\theta$$

is valid when θ is measured in **radians**.

Do not directly put degree measure into this formula.

10. Example Based on Arc Length**SOLVED EXAMPLE**

Find the length of an arc of a circle of radius 7 cm subtending an angle $\frac{\pi}{3}$ at the centre.
Given,

$$r = 7 \text{ cm}$$

and

$$\theta = \frac{\pi}{3}.$$

Using

$$s = r\theta,$$

we get

$$s = 7 \times \frac{\pi}{3}.$$

Therefore,

$$s = \frac{7\pi}{3} \text{ cm}.$$

11. Area of a Sector

If a sector of a circle has radius r and angle θ radians, then

KEY FORMULA

$$A = \frac{1}{2}r^2\theta$$

SOLVED EXAMPLE

Find the area of a sector having radius 6 cm and angle $\frac{\pi}{3}$.

$$\begin{aligned}A &= \frac{1}{2}r^2\theta \\&= \frac{1}{2}(6)^2\left(\frac{\pi}{3}\right) \\&= 18\pi.\end{aligned}$$

Hence,

$$\boxed{A = 18\pi \text{ cm}^2}.$$

12. Important Results

KEY FORMULA

$$\boxed{360^\circ = 2\pi \text{ rad}}$$

$$\boxed{180^\circ = \pi \text{ rad}}$$

$$\boxed{1^\circ = \frac{\pi}{180} \text{ rad}}$$

$$\boxed{1 \text{ rad} = \frac{180^\circ}{\pi}}$$

$$\boxed{s = r\theta}$$

$$\boxed{A = \frac{1}{2}r^2\theta}$$

13. Exam Tricks

EXAM TRICK

1. Whenever you see π , think about **radian measure**.

2. Whenever you see 180° , immediately remember

$$180^\circ = \pi.$$

3. For degree $\rightarrow$ radian, multiply by

$$\frac{\pi}{180}.$$

4. For radian $\rightarrow$ degree, multiply by

$$\frac{180}{\pi}.$$

5. In the formula $s = r\theta$, always use θ in radians.

6. In the formula $A = \frac{1}{2}r^2\theta$, always use θ in radians.

14. Practice Questions

1. Convert 45° into radians.
2. Convert 135° into radians.
3. Convert 240° into radians.
4. Convert $\frac{2\pi}{3}$ radians into degrees.
5. Convert $\frac{7\pi}{6}$ radians into degrees.
6. Find the length of an arc of radius 14 cm subtending an angle $\frac{\pi}{4}$.
7. Find the area of a sector of radius 10 cm and angle $\frac{\pi}{6}$.
8. An arc of a circle has length 22 cm and radius 7 cm. Find the angle subtended at the centre in radians.

15. Quick Revision

ONE-MINUTE REVISION

$$180^\circ = \pi \text{ rad}$$

$$\text{Degree} \rightarrow \text{Radian} : \quad \times \frac{\pi}{180}$$

$$\text{Radian} \rightarrow \text{Degree} : \quad \times \frac{180}{\pi}$$

$$s = r\theta$$

$$A = \frac{1}{2}r^2\theta$$

Ravi Shankar – Mathematics

Class 11 — Trigonometry — Important Notes

PART 2: TRIGONOMETRIC FUNCTIONS

1. Introduction to Trigonometric Functions

Trigonometry trigonometric functions :

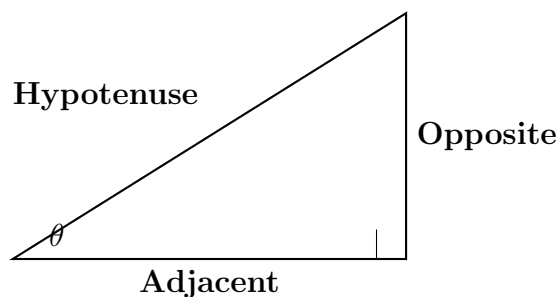
$$\sin \theta, \quad \cos \theta, \quad \tan \theta, \quad \cot \theta, \quad \sec \theta, \quad \csc \theta$$

$\sin \theta, \cos \theta, \tan \theta$ **primary functions**

$\cot \theta, \sec \theta, \csc \theta$ reciprocal functions

2. Trigonometric Ratios in a Right Triangle

right-angled triangle θ



θ respect :

Opposite side = Perpendicular

Adjacent side = Base

Hypotenuse = side

3. Six Trigonometric Ratios

KEY FORMULA

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Perpendicular}}{\text{Base}}$$

$$\cot \theta = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\sec \theta = \frac{\text{Hypotenuse}}{\text{Base}}$$

$$\csc \theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}}$$

4. Reciprocal Relations

KEY FORMULA

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

5. Quotient Relations

KEY FORMULA

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

relations important identity questions solve

6. Cartesian Coordinate System

angle coordinate plane represent ,

$$P(x, y)$$

$$OP = r.$$

$$r = \sqrt{x^2 + y^2}$$

trigonometric functions :

KEY FORMULA

$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x}$$

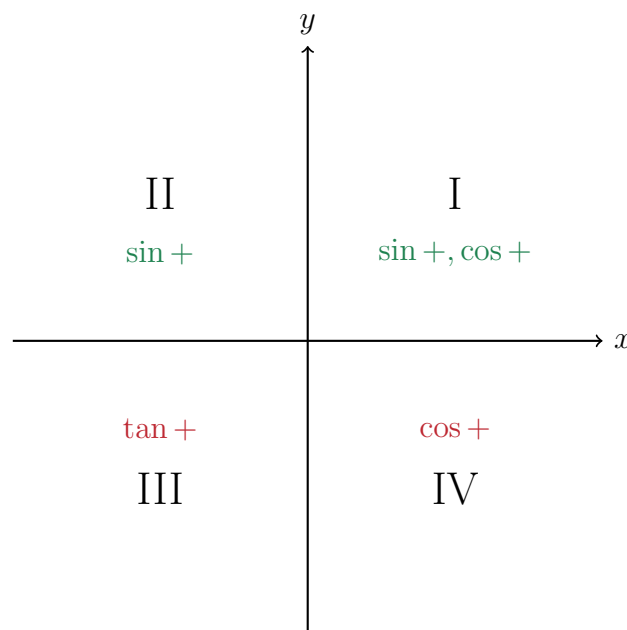
$$\cot \theta = \frac{x}{y}$$

$$\sec \theta = \frac{r}{x}$$

$$\csc \theta = \frac{r}{y}$$

7. Signs of Trigonometric Functions

Coordinate plane quadrants divided



8. ASTC Rule

IMPORTANT

Trigonometric functions signs

ASTC

rule important

A – All
S – Sine
T – Tangent
C – Cosine

Quadrant I
 functions positive

$$\sin \theta > 0, \quad \cos \theta > 0, \quad \tan \theta > 0$$

Quadrant II
 sine positive

$$\sin \theta > 0$$

Quadrant III
 tangent positive

$$\tan \theta > 0$$

Quadrant IV
 cosine positive

$$\cos \theta > 0$$

9. Sign Table

	I	II	III	IV
$\sin \theta$	+	+	–	–
$\cos \theta$	+	–	–	+
$\tan \theta$	+	–	+	–
$\cot \theta$	+	–	+	–
$\sec \theta$	+	–	–	+
$\csc \theta$	+	+	–	–

10. Standard Values

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined

11. Standard Values Trick

EXAM TRICK

$\sin \theta$ sequence:

$$\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$$

$$0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$$

$\cos \theta$ sequence reverse

$$1, \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}, 0$$

12. Fundamental Trigonometric Identities

important identities:

KEY FORMULA

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

identities **Pythagorean Identities**

13. Derived Identities

identity:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$\cos^2 \theta$ divide ,

$$\frac{\sin^2 \theta}{\cos^2 \theta} + 1 = \frac{1}{\cos^2 \theta}$$

$$\boxed{1 + \tan^2 \theta = \sec^2 \theta}.$$

$\sin^2 \theta$ divide ,

$$\boxed{1 + \cot^2 \theta = \csc^2 \theta}.$$

14. Important Example

SOLVED EXAMPLE

$$\sin \theta = \frac{3}{5}$$

θ acute angle , $\cos \theta$ $\tan \theta$

,

$$\sin^2 \theta + \cos^2 \theta = 1.$$

$$\left(\frac{3}{5}\right)^2 + \cos^2 \theta = 1$$

$$\frac{9}{25} + \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{16}{25}.$$

θ acute ,

$$\boxed{\cos \theta = \frac{4}{5}}.$$

,

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{3/5}{4/5}$$

$$\boxed{\tan \theta = \frac{3}{4}}.$$

15. Domain and Range

Function	Domain	Range
$\sin x$	$\mathbb{R}$	$[-1, 1]$
$\cos x$	$\mathbb{R}$	$[-1, 1]$
$\tan x$	$\mathbb{R} \setminus \left\{ \frac{(2n+1)\pi}{2} : n \in \mathbb{Z} \right\}$	$\mathbb{R}$
$\cot x$	$\mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$	$\mathbb{R}$
$\sec x$	$\mathbb{R} \setminus \left\{ \frac{(2n+1)\pi}{2} : n \in \mathbb{Z} \right\}$	$(-\infty, -1] \cup [1, \infty)$
$\csc x$	$\mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$	$(-\infty, -1] \cup [1, \infty)$

16. Important Exam Points

IMPORTANT

1. $\sin \theta$ $\cos \theta$ values $[-1, 1]$
2. $\tan 90^\circ$ defined
3. $\cot 0^\circ$ defined
4. $\sec 90^\circ$ defined
5. $\csc 0^\circ$ defined
6. $\tan \theta$ range real line
7. $\sec \theta$ $\csc \theta$ absolute value 1

17. Most Important Formula Sheet

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

18. Important Practice Questions

1. If $\sin \theta = \frac{5}{13}$ and θ is acute, find $\cos \theta$ and $\tan \theta$.
2. If $\cos \theta = -\frac{12}{13}$ and θ lies in the second quadrant, find $\sin \theta$ and $\tan \theta$.
3. Find the value of

$$\sin 30^\circ + \cos 60^\circ.$$

4. Find the value of

$$\tan 45^\circ + \sin 90^\circ.$$

5. Prove that

$$\frac{1 - \cos^2 \theta}{\sin^2 \theta} = 1.$$

6. Prove that

$$\frac{\sec^2 \theta - 1}{\tan^2 \theta} = 1.$$

7. Find the range of $\sin x$.
8. Find the domain of $\tan x$.

FINAL REVISION

TRIGONOMETRY = RATIOS + SIGNS + IDENTITIES

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

PART 3: COMPOUND ANGLES

1. Introduction

angles A and B are **Compound Angle**
if

$$A + B, \quad A - B$$

Trigonometry identities :

KEY FORMULA

$$\sin(A + B)$$

$$\sin(A - B)$$

$$\cos(A + B)$$

$$\cos(A - B)$$

$$\tan(A + B)$$

$$\tan(A - B)$$

2. Sine of Sum of Two Angles

identity:

KEY FORMULA

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

EXAM TRICK

trick:

$$\sin(A + B) = SC + CS$$

$$\sin A \cos B + \cos A \sin B.$$

SOLVED EXAMPLE

Find:

$$\sin 75^\circ.$$

:

$$75^\circ = 45^\circ + 30^\circ.$$

$$\sin 75^\circ = \sin(45^\circ + 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ.$$

standard values ,

$$\begin{aligned} &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}}. \end{aligned}$$

Rationalising,

$$\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

3. Sine of Difference of Two Angles**KEY FORMULA**

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

EXAM TRICK

:

$$\sin(A + B) = SC + CS$$

$$\sin(A - B) = SC - CS$$

sine sign bracket

SOLVED EXAMPLE

Find:

$$\sin 15^\circ.$$

$$15^\circ = 45^\circ - 30^\circ,$$

$$\sin 15^\circ = \sin(45^\circ - 30^\circ).$$

$$= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ.$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}.$$

Therefore,

$$\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$$

4. Cosine of Sum of Two Angles**KEY FORMULA**

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

EXAM TRICKCosine sign **opposite**

$$\cos(A + B) = CC - SS$$

SOLVED EXAMPLE

Find:

$$\cos 75^\circ.$$

$$75^\circ = 45^\circ + 30^\circ.$$

Therefore,

$$\begin{aligned}\cos 75^\circ &= \cos(45^\circ + 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ.\end{aligned}$$

$$= \frac{1}{\sqrt{2}} \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \frac{1}{2}.$$

Hence,

$$\boxed{\cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}}$$

5. Cosine of Difference of Two Angles**KEY FORMULA**

$$\boxed{\cos(A - B) = \cos A \cos B + \sin A \sin B}$$

EXAM TRICK

:

$$\boxed{\cos(A - B) = CC + SS}$$

cosine bracket minus sign **plus**

SOLVED EXAMPLE

Find:

$$\cos 15^\circ.$$

$$15^\circ = 45^\circ - 30^\circ.$$

Hence,

$$\cos 15^\circ = \cos(45^\circ - 30^\circ)$$

$$= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ.$$

Thus,

$$\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

6. Tangent of Sum of Two Angles**KEY FORMULA**

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

IMPORTANT

formula valid denominator zero

$$1 - \tan A \tan B \neq 0.$$

7. Tangent of Difference of Two Angles**KEY FORMULA**

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

EXAM TRICK

formulas :

$$\tan(A + B) = \frac{T_A + T_B}{1 - T_A T_B}$$

$$\tan(A - B) = \frac{T_A - T_B}{1 + T_A T_B}$$

Numerator denominator sign

8. Complete Compound Angle Formula Sheet

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

9. Important Values Using Compound Angles**Value of $\sin 75^\circ$**

$$75^\circ = 45^\circ + 30^\circ$$

$$\sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

$$\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Value of $\cos 75^\circ$

$$\cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$$

Value of $\sin 15^\circ$

$$\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$$

Value of $\cos 15^\circ$

$$\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

10. Finding $\tan 15^\circ$ **SOLVED EXAMPLE**

$$15^\circ = 45^\circ - 30^\circ.$$

Therefore,

$$\tan 15^\circ = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}.$$

Using

$$\tan 45^\circ = 1, \quad \tan 30^\circ = \frac{1}{\sqrt{3}},$$

we get

$$\tan 15^\circ = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}}.$$

Multiplying numerator and denominator by $\sqrt{3}$,

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}.$$

Rationalising,

$$\begin{aligned} &= \frac{(\sqrt{3} - 1)^2}{3 - 1} \\ &= \frac{4 - 2\sqrt{3}}{2}. \end{aligned}$$

Hence,

$$\tan 15^\circ = 2 - \sqrt{3}$$

11. Finding $\tan 75^\circ$

SOLVED EXAMPLE

$$75^\circ = 45^\circ + 30^\circ.$$

Therefore,

$$\tan 75^\circ = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}.$$

Multiplying numerator and denominator by $\sqrt{3}$,

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}.$$

Rationalising,

$$\begin{aligned} &= \frac{(\sqrt{3} + 1)^2}{2} \\ &= \frac{4 + 2\sqrt{3}}{2}. \end{aligned}$$

Hence,

$$\boxed{\tan 75^\circ = 2 + \sqrt{3}}$$

12. Important Special Results

KEY FORMULA

$$\boxed{\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}}$$

$$\boxed{\cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}}$$

$$\boxed{\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}}$$

$$\boxed{\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}}$$

$$\boxed{\tan 15^\circ = 2 - \sqrt{3}}$$

$$\boxed{\tan 75^\circ = 2 + \sqrt{3}}$$

13. Important Identity Based Question

SOLVED EXAMPLE

Prove that

$$\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B.$$

Using compound angle formulas,

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

and

$$\sin(A - B) = \sin A \cos B - \cos A \sin B.$$

Therefore,

$$\begin{aligned} & \sin(A + B) \sin(A - B) \\ &= (\sin A \cos B + \cos A \sin B)(\sin A \cos B - \cos A \sin B). \end{aligned}$$

Using

$$(x + y)(x - y) = x^2 - y^2,$$

we get

$$= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B.$$

Now,

$$\cos^2 B = 1 - \sin^2 B$$

and

$$\cos^2 A = 1 - \sin^2 A.$$

Therefore,

$$= \sin^2 A(1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B.$$

On simplifying,

$$\boxed{\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B}$$

14. Important Exam Tricks

EXAM TRICK

Trick 1: Sine

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Sine sign **same**

Trick 2: Cosine

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Cosine sign **opposite**

Trick 3: Tangent

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

15. Important Practice Questions

- Find $\sin 75^\circ$.
- Find $\cos 75^\circ$.
- Find $\sin 15^\circ$.
- Find $\cos 15^\circ$.
- Find $\tan 15^\circ$.
- Find $\tan 75^\circ$.
- Prove that

$$\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B.$$

- Prove that

$$\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B.$$

- If

$$\tan A = \frac{1}{2}, \quad \tan B = \frac{1}{3},$$

find $\tan(A + B)$.

- If

$$\tan A = \frac{3}{4}, \quad \tan B = \frac{1}{4},$$

find $\tan(A - B)$.

11. Prove that

$$\frac{\sin(A+B)}{\sin(A-B)} = \frac{\tan A + \tan B}{\tan A - \tan B}.$$

16. One-Page Revision

COMPOUND ANGLES

$$\sin(A+B) = SC + CS$$

$$\sin(A-B) = SC - CS$$

$$\cos(A+B) = CC - SS$$

$$\cos(A-B) = CC + SS$$

$$\tan(A+B) = \frac{T_A + T_B}{1 - T_A T_B}$$

$$\tan(A-B) = \frac{T_A - T_B}{1 + T_A T_B}$$

$$\tan 15^\circ = 2 - \sqrt{3}$$

$$\tan 75^\circ = 2 + \sqrt{3}$$

PART 4: MULTIPLE & SUB-MULTIPLE ANGLES

1. Introduction

angle integer multiply , **Multiple Angle**
:

$$2A, \quad 3A, \quad 4A, \quad \dots$$

angle integer divide , **Sub-multiple Angle**
:

$$\frac{A}{2}, \quad \frac{A}{3}, \quad \frac{A}{4}.$$

Class 11 important :

$$2A, \quad 3A, \quad \frac{A}{2}$$

2. Double Angle Formula for Sine

Compound angle formula ,

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

$$B = A ,$$

$$\sin(A + A) = \sin A \cos A + \cos A \sin A.$$

$$\boxed{\sin 2A = 2 \sin A \cos A}$$

KEY FORMULA

$$\boxed{\sin 2A = 2 \sin A \cos A}$$

3. Double Angle Formula for Cosine

,

$$\cos(A + B) = \cos A \cos B - \sin A \sin B.$$

$$B = A ,$$

$$\cos 2A = \cos^2 A - \sin^2 A.$$

KEY FORMULA

$$\boxed{\cos 2A = \cos^2 A - \sin^2 A}$$

$$\sin^2 A + \cos^2 A = 1$$

forms

$$\cos^2 A = 1 - \sin^2 A$$

$$\boxed{\cos 2A = 1 - 2 \sin^2 A}$$

,

$$\sin^2 A = 1 - \cos^2 A$$

$$\boxed{\cos 2A = 2 \cos^2 A - 1}$$

important forms:

KEY FORMULA

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A = 2\cos^2 A - 1$$

4. Double Angle Formula for Tangent

Compound angle formula ,

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

$$B = A ,$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}.$$

Therefore,

KEY FORMULA

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

denominator zero

5. Complete Double Angle Formula Sheet

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A = 2\cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

6. Expressing $\sin^2 A$ and $\cos^2 A$

From

$$\cos 2A = 1 - 2\sin^2 A,$$

we get

$$2 \sin^2 A = 1 - \cos 2A.$$

Therefore,

KEY FORMULA

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

Similarly,

$$\cos 2A = 2 \cos^2 A - 1$$

gives

$$2 \cos^2 A = 1 + \cos 2A.$$

Therefore,

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

7. Important Example

SOLVED EXAMPLE

If

$$\sin A = \frac{3}{5}$$

and A is acute, find $\sin 2A$ and $\cos 2A$.

Since A is acute,

$$\cos A = \frac{4}{5}.$$

Therefore,

$$\sin 2A = 2 \sin A \cos A$$

$$= 2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right)$$

$$\boxed{\sin 2A = \frac{24}{25}}$$

Now,

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= \frac{16}{25} - \frac{9}{25}$$

$$\boxed{\cos 2A = \frac{7}{25}}$$

8. Triple Angle Formula for Sine

,

$$\sin 3A = \sin(2A + A).$$

Using compound angle formula,

$$\sin 3A = \sin 2A \cos A + \cos 2A \sin A.$$

,

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A.$$

$$\sin 3A = 2 \sin A \cos^2 A + (\cos^2 A - \sin^2 A) \sin A.$$

Simplifying,

$$\sin 3A = 3 \sin A \cos^2 A - \sin^3 A.$$

$$\cos^2 A = 1 - \sin^2 A.$$

,

$$\sin 3A = 3 \sin A(1 - \sin^2 A) - \sin^3 A.$$

Hence,

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

9. Triple Angle Formula for Cosine

Similarly,

$$\cos 3A = \cos(2A + A).$$

Using

$$\cos(A + B) = \cos A \cos B - \sin A \sin B,$$

we obtain

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

useful form:

$$\cos 3A = \cos A(4 \cos^2 A - 3)$$

10. Triple Angle Formula for Tangent

Using

$$\tan 3A = \tan(2A + A),$$

we obtain

$$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

11. Triple Angle Formula Sheet

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

12. Half Angle Formula

Double-angle formulas half-angle formulas

,

$$\cos 2A = 1 - 2 \sin^2 A.$$

$$2A = \theta,$$

$$\cos \theta = 1 - 2 \sin^2 \frac{\theta}{2}.$$

,

$$2 \sin^2 \frac{\theta}{2} = 1 - \cos \theta.$$

Hence,

KEY FORMULA

$$\sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2}$$

,

$$\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1.$$

Therefore,

$$\cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2}$$

13. Half Angle Formula with Sign

Square root ,

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

Sign depend

$$\frac{\theta}{2}$$

quadrant

IMPORTANT

Important:

Half-angle formulas $\pm$ ignore

Quadrant sign

14. Formula for $\tan \frac{\theta}{2}$

,

$$\tan \frac{\theta}{2} = \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}.$$

,

KEY FORMULA

$$\tan \frac{\theta}{2} = \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

sign quadrant

useful form:

$$\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta}$$

15. Important Example: Half Angle

SOLVED EXAMPLE

If

$$\cos A = \frac{3}{5}$$

and A lies in the first quadrant, find

$$\sin \frac{A}{2}.$$

Using,

$$\sin \frac{A}{2} = \sqrt{\frac{1 - \cos A}{2}}.$$

Therefore,

$$= \sqrt{\frac{1 - \frac{3}{5}}{2}}$$

$$= \sqrt{\frac{\frac{2}{5}}{2}}$$

$$= \sqrt{\frac{1}{5}}.$$

Hence,

$$\boxed{\sin \frac{A}{2} = \frac{1}{\sqrt{5}}}$$

16. Important Example: $\cos \frac{A}{2}$

SOLVED EXAMPLE

If

$$\cos A = \frac{3}{5}$$

and A is acute, find

$$\cos \frac{A}{2}.$$

Using,

$$\cos \frac{A}{2} = \sqrt{\frac{1 + \cos A}{2}}.$$

Thus,

$$= \sqrt{\frac{1 + \frac{3}{5}}{2}}$$

$$= \sqrt{\frac{\frac{8}{5}}{2}}$$

$$= \sqrt{\frac{4}{5}}.$$

Hence,

$$\boxed{\cos \frac{A}{2} = \frac{2}{\sqrt{5}}}$$

17. Important Product Forms

Double-angle formulas :

$$\sin 2A = 2 \sin A \cos A.$$

Therefore,

KEY FORMULA

$$\boxed{2 \sin A \cos A = \sin 2A}$$

Similarly,

$$\boxed{\cos^2 A - \sin^2 A = \cos 2A}$$

$$\boxed{2 \cos^2 A = 1 + \cos 2A}$$

$$\boxed{2 \sin^2 A = 1 - \cos 2A}$$

18. Important Exam Question

SOLVED EXAMPLE

Prove that

$$\frac{1 - \cos 2A}{\sin 2A} = \tan A.$$

Using

$$1 - \cos 2A = 2 \sin^2 A$$

and

$$\sin 2A = 2 \sin A \cos A,$$

we get

$$\frac{1 - \cos 2A}{\sin 2A} = \frac{2 \sin^2 A}{2 \sin A \cos A}.$$

Canceling common terms,

$$= \frac{\sin A}{\cos A}.$$

Therefore,

$$\boxed{\frac{1 - \cos 2A}{\sin 2A} = \tan A}$$

19. Important Exam Question

SOLVED EXAMPLE

Prove that

$$\frac{\sin 2A}{1 + \cos 2A} = \tan A.$$

We know,

$$\sin 2A = 2 \sin A \cos A$$

and

$$1 + \cos 2A = 2 \cos^2 A.$$

Therefore,

$$\frac{\sin 2A}{1 + \cos 2A} = \frac{2 \sin A \cos A}{2 \cos^2 A}.$$

Hence,

$$= \frac{\sin A}{\cos A}.$$

Therefore,

$$\boxed{\frac{\sin 2A}{1 + \cos 2A} = \tan A}$$

20. Must Remember

IMPORTANT

1.

$$\sin 2A = 2 \sin A \cos A$$

2.

$$\cos 2A = 1 - 2 \sin^2 A$$

3.

$$\cos 2A = 2 \cos^2 A - 1$$

4.

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

5.

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

6.

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

7.

$$\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$$

8.

$$\cos^2 \frac{A}{2} = \frac{1 + \cos A}{2}$$

21. Important Practice Questions

1. If $\sin A = \frac{3}{5}$, find $\sin 2A$ and $\cos 2A$.
2. If $\cos A = \frac{5}{13}$, find $\sin 2A$.
3. Find $\sin 30^\circ$ using the double-angle formula.
4. Find $\cos 60^\circ$ using the double-angle formula.
5. Find $\sin 15^\circ$ using the half-angle formula.
6. Find $\cos 15^\circ$ using the half-angle formula.
7. Prove that

$$\frac{1 - \cos 2A}{\sin 2A} = \tan A.$$

8. Prove that

$$\frac{\sin 2A}{1 + \cos 2A} = \tan A.$$

9. Prove that

$$\frac{1 - \cos 2A}{1 + \cos 2A} = \tan^2 A.$$

10. If

$$\tan A = \frac{1}{2},$$

find $\tan 2A$.

11. If

$$\sin A = \frac{4}{5},$$

find $\cos 2A$.

12. Prove that

$$\sin 3A = 3 \sin A - 4 \sin^3 A.$$

13. Prove that

$$\cos 3A = 4 \cos^3 A - 3 \cos A.$$

22. One-Minute Revision

MULTIPLE & SUB-MULTIPLE ANGLES

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = 1 - 2 \sin^2 A$$

$$\cos 2A = 2 \cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$$

$$\cos^2 \frac{A}{2} = \frac{1 + \cos A}{2}$$

$$\tan \frac{A}{2} = \frac{\sin A}{1 + \cos A} = \frac{1 - \cos A}{\sin A}$$

PART 5: TRANSFORMATION FORMULAE

1. Introduction

Trigonometric expressions form form **Transformation Formulae**
transformations :

$$\text{Sum/Difference} \longleftrightarrow \text{Product}$$

1. Sum-to-Product Formulae

2. Product-to-Sum Formulae

expressions simplify identities prove

2. Sum-to-Product Formulae

angles A B
important formulas:

KEY FORMULA

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

3. Easy Memory Trick

EXAM TRICK

Sine + Sine

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

:

$$S + S = 2SC$$

—

Sine - Sine

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

:

$$S - S = 2CS$$

—

Cosine + Cosine

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

:

$$C + C = 2CC$$

—

Cosine - Cosine

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

:

$$C - C = -2SS$$

4. Product-to-Sum Formulae

product sum formulas:

KEY FORMULA

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

5. Complete Transformation Formula Sheet**Sum $\rightarrow$ Product**

$$\sin A + \sin B = 2 \sin \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A + B}{2} \sin \frac{A - B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A + B}{2} \sin \frac{A - B}{2}$$

Product $\rightarrow$ Sum

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

6. Derivation of $\sin A + \sin B$

,

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B.$$

equations add ,

$$\sin(A + B) + \sin(A - B) = 2 \sin A \cos B.$$

suitable substitution sum-to-product form :

$$\sin A + \sin B = 2 \sin \frac{A + B}{2} \cos \frac{A - B}{2}$$

7. Example: Sum-to-Product

SOLVED EXAMPLE

Simplify:

$$\sin 70^\circ + \sin 10^\circ.$$

Using

$$\sin A + \sin B = 2 \sin \frac{A + B}{2} \cos \frac{A - B}{2},$$

we get

$$= 2 \sin \frac{70^\circ + 10^\circ}{2} \cos \frac{70^\circ - 10^\circ}{2}.$$

Therefore,

$$= 2 \sin 40^\circ \cos 30^\circ.$$

Since

$$\cos 30^\circ = \frac{\sqrt{3}}{2},$$

we get

$$\sin 70^\circ + \sin 10^\circ = \sqrt{3} \sin 40^\circ$$

8. Example: Difference of Sines

SOLVED EXAMPLE

Simplify:

$$\sin 80^\circ - \sin 20^\circ.$$

Using

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2},$$

we get

$$= 2 \cos \frac{80^\circ + 20^\circ}{2} \sin \frac{80^\circ - 20^\circ}{2}.$$

Thus,

$$= 2 \cos 50^\circ \sin 30^\circ.$$

Since

$$\sin 30^\circ = \frac{1}{2},$$

therefore

$$\boxed{\sin 80^\circ - \sin 20^\circ = \cos 50^\circ}$$

9. Example: Sum of Cosines

SOLVED EXAMPLE

Simplify:

$$\cos 70^\circ + \cos 10^\circ.$$

Using

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2},$$

we obtain

$$= 2 \cos 40^\circ \cos 30^\circ.$$

Therefore,

$$\boxed{\cos 70^\circ + \cos 10^\circ = \sqrt{3} \cos 40^\circ}$$

10. Example: Difference of Cosines

SOLVED EXAMPLE

Simplify:

$$\cos 80^\circ - \cos 20^\circ.$$

Using

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2},$$

we get

$$= -2 \sin 50^\circ \sin 30^\circ.$$

Since

$$\sin 30^\circ = \frac{1}{2},$$

we obtain

$$\boxed{\cos 80^\circ - \cos 20^\circ = -\sin 50^\circ}$$

11. Product-to-Sum Example

SOLVED EXAMPLE

Express

$$2 \sin 50^\circ \cos 20^\circ$$

as a sum.

Using

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B),$$

we get

$$= \sin 70^\circ + \sin 30^\circ.$$

Therefore,

$$\boxed{2 \sin 50^\circ \cos 20^\circ = \sin 70^\circ + \frac{1}{2}}$$

12. Product of Two Cosines

SOLVED EXAMPLE

Express

$$2 \cos 50^\circ \cos 10^\circ$$

as a sum.

Using

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B),$$

we obtain

$$= \cos 60^\circ + \cos 40^\circ.$$

Hence,

$$2 \cos 50^\circ \cos 10^\circ = \frac{1}{2} + \cos 40^\circ$$

13. Product of Two Sines

SOLVED EXAMPLE

Express

$$2 \sin 40^\circ \sin 20^\circ$$

as a sum/difference.

Using

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B),$$

we get

$$= \cos 20^\circ - \cos 60^\circ.$$

Therefore,

$$2 \sin 40^\circ \sin 20^\circ = \cos 20^\circ - \frac{1}{2}$$

14. Important Identity

Prove that

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}.$$

Put

$$x = \frac{A+B}{2}, \quad y = \frac{A-B}{2}.$$

Then

$$A = x + y$$

and

$$B = x - y.$$

Therefore,

$$\sin A + \sin B = \sin(x + y) + \sin(x - y).$$

Using compound-angle formula,

$$= (\sin x \cos y + \cos x \sin y) + (\sin x \cos y - \cos x \sin y).$$

Hence,

$$= 2 \sin x \cos y.$$

Substituting back,

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

15. Useful Special Transformations

expressions transform

KEY FORMULA

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

16. Exam Strategy

EXAM TRICK

question

$$\sin A + \sin B$$

$$\cos A + \cos B$$

expression , **Sum-to-Product**
question

$$2 \sin A \cos B$$

$$2 \cos A \cos B$$

expression , **Product-to-Sum**

17. Important Practice Questions

1. Simplify:

$$\sin 70^\circ + \sin 10^\circ.$$

2. Simplify:

$$\sin 80^\circ - \sin 20^\circ.$$

3. Simplify:

$$\cos 70^\circ + \cos 10^\circ.$$

4. Simplify:

$$\cos 80^\circ - \cos 20^\circ.$$

5. Express as a sum:

$$2 \sin 50^\circ \cos 20^\circ.$$

6. Express as a sum:

$$2 \cos 50^\circ \cos 10^\circ.$$

7. Express as a difference:

$$2 \sin 40^\circ \sin 20^\circ.$$

8. Prove:

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}.$$

9. Prove:

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}.$$

10. Prove:

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}.$$

11. Prove:

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B).$$

12. Prove:

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B).$$

18. One-Minute Revision

TRANSFORMATION FORMULAE

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

PART 6: TRIGONOMETRIC EQUATIONS

1. Introduction

equation trigonometric functions , **Trigonometric Equation**
:

$$\sin x = \frac{1}{2}$$

$$\cos x = 0$$

$$\tan x = 1$$

equations solutions trigonometric functions periodic

2. Periodicity

Trigonometric functions fixed intervals values repeat

KEY FORMULA

$\sin(x + 2\pi) = \sin x$

$\cos(x + 2\pi) = \cos x$

$\tan(x + \pi) = \tan x$

:

Period of $\sin x = 2\pi$

Period of $\cos x = 2\pi$

Period of $\tan x = \pi$

3. General Solution of $\sin x = \sin \alpha$

$$\sin x = \sin \alpha,$$

general solution :

KEY FORMULA

$x = n\pi + (-1)^n \alpha, \qquad n \in \mathbb{Z}$

equivalent forms :

$x = 2n\pi + \alpha$

$$x = (2n + 1)\pi - \alpha$$

$$n \in \mathbb{Z}.$$

4. Why Two Families?

Sine function complete revolution generally angles same sine value

$$\sin x = \sin \alpha,$$

$$x = \alpha$$

$$x = \pi - \alpha.$$

periodicity $2n\pi$ add
:

$$x = 2n\pi + \alpha$$

$$x = 2n\pi + \pi - \alpha.$$

5. Example: $\sin x = \frac{1}{2}$

SOLVED EXAMPLE

Solve:

$$\sin x = \frac{1}{2}.$$

:

$$\sin 30^\circ = \frac{1}{2}.$$

$$\alpha = 30^\circ.$$

Sine positive Quadrant I II

,

$$x = 30^\circ$$

$$x = 180^\circ - 30^\circ = 150^\circ.$$

General solution:

$$x = 360^\circ n + 30^\circ$$

$$x = 360^\circ n + 150^\circ$$

$$n \in \mathbb{Z}.$$

Radians :

$$x = 2n\pi + \frac{\pi}{6}$$

$$x = 2n\pi + \frac{5\pi}{6}$$

6. General Solution of $\cos x = \cos \alpha$

$$\cos x = \cos \alpha,$$

general solution:

KEY FORMULA

$$x = 2n\pi \pm \alpha, \quad n \in \mathbb{Z}$$

,

$$x = 2n\pi + \alpha$$

$$x = 2n\pi - \alpha.$$

7. Example: $\cos x = \frac{1}{2}$

SOLVED EXAMPLE

Solve:

$$\cos x = \frac{1}{2}.$$

:

$$\cos 60^\circ = \frac{1}{2}.$$

Cosine positive Quadrant I IV

,

$$x = 60^\circ$$

$$x = 360^\circ - 60^\circ = 300^\circ.$$

General solution:

$$x = 360^\circ n + 60^\circ$$

$$x = 360^\circ n + 300^\circ$$

Radians :

$$x = 2n\pi + \frac{\pi}{3}$$

$$x = 2n\pi - \frac{\pi}{3}$$

8. General Solution of $\tan x = \tan \alpha$

$$\tan x = \tan \alpha,$$

tangent period π

:

KEY FORMULA

$$x = n\pi + \alpha, \quad n \in \mathbb{Z}$$

9. Example: $\tan x = 1$

SOLVED EXAMPLE

Solve:

$$\tan x = 1.$$

:

$$\tan 45^\circ = 1.$$

Therefore,

$$\alpha = 45^\circ.$$

General solution:

$$x = 45^\circ + 180^\circ n$$

radians :

$$x = \frac{\pi}{4} + n\pi$$

$$n \in \mathbb{Z}.$$

10. General Solution Formula Sheet

$$\sin x = \sin \alpha \Rightarrow x = n\pi + (-1)^n \alpha$$

$$\cos x = \cos \alpha \Rightarrow x = 2n\pi \pm \alpha$$

$$\tan x = \tan \alpha \Rightarrow x = n\pi + \alpha$$

11. Important Standard Equations

KEY FORMULA

$$\sin x = 0 \Rightarrow x = n\pi$$

$$\cos x = 0 \Rightarrow x = \frac{(2n+1)\pi}{2}$$

$$\tan x = 0 \Rightarrow x = n\pi$$

12. Equation $\sin x = 0$

$$\sin x = 0.$$

Sine zero :

$$x = 0, \pi, 2\pi, 3\pi, \dots$$

Therefore,

$$\boxed{x = n\pi, \quad n \in \mathbb{Z}.}$$

13. Equation $\cos x = 0$

$$\cos x = 0.$$

Cosine zero :

$$x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

Therefore,

$$\boxed{x = \frac{(2n+1)\pi}{2}, \quad n \in \mathbb{Z}.}$$

14. Equation $\tan x = 0$

$$\tan x = 0.$$

Tangent zero :

$$x = 0, \pi, 2\pi, \dots$$

Therefore,

$$\boxed{x = n\pi, \quad n \in \mathbb{Z}.}$$

15. Equations of the Form $\sin x = \pm 1$

For

$$\sin x = 1,$$

$$\boxed{x = \frac{\pi}{2} + 2n\pi}$$

For

$$\sin x = -1,$$

$$\boxed{x = \frac{3\pi}{2} + 2n\pi}$$

16. Equations of the Form $\cos x = \pm 1$

For

$$\cos x = 1,$$

$$\boxed{x = 2n\pi}$$

For

$$\cos x = -1,$$

$$\boxed{x = (2n + 1)\pi}$$

17. Example with an Interval

SOLVED EXAMPLE

Solve

$$\sin x = \frac{1}{2},$$

for

$$0 \leq x \leq 2\pi.$$

We know:

$$\sin x = \frac{1}{2}$$

at

$$x = \frac{\pi}{6}$$

and

$$x = \frac{5\pi}{6}.$$

Therefore,

$$\boxed{x = \frac{\pi}{6}, \frac{5\pi}{6}}$$

are the required solutions.

18. Example: $\cos x = -\frac{1}{2}$ **SOLVED EXAMPLE**

Solve

$$\cos x = -\frac{1}{2}, \quad 0 \leq x \leq 2\pi.$$

We know:

$$\cos 60^\circ = \frac{1}{2}.$$

Cosine negative Quadrant II III

Therefore,

$$x = 120^\circ$$

$$x = 240^\circ.$$

Radians :

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

19. Example: $\tan x = -1$ **SOLVED EXAMPLE**

Solve

$$\tan x = -1, \quad 0 \leq x \leq 2\pi.$$

We know:

$$\tan 45^\circ = 1.$$

Tangent negative Quadrant II IV

Therefore,

$$x = 135^\circ$$

$$x = 315^\circ.$$

Hence,

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

20. Equation $\sin^2 x = \sin^2 \alpha$

$$\sin^2 x = \sin^2 \alpha,$$

$$\sin x = \pm \sin \alpha.$$

convenient general solution :

KEY FORMULA

$$x = n\pi \pm \alpha, \quad n \in \mathbb{Z}$$

21. Equation $\cos^2 x = \cos^2 \alpha$

,

$$\cos^2 x = \cos^2 \alpha$$

$$\cos x = \pm \cos \alpha.$$

General solution:

$$x = n\pi \pm \alpha, \quad n \in \mathbb{Z}.$$

22. Equation $\tan^2 x = \tan^2 \alpha$

$$\tan^2 x = \tan^2 \alpha,$$

$$\tan x = \pm \tan \alpha.$$

,

$$x = n\pi \pm \alpha, \quad n \in \mathbb{Z}.$$

23. Important Method for Trigonometric Equations

IMPORTANT

trigonometric equation solve :

1. Equation simplest form
2. possible trigonometric function
3. Standard value identify
4. Quadrants signs
5. Required interval solutions
6. general solution , $n \in \mathbb{Z}$

24. Exam Trick: CAST / ASTC

EXAM TRICK

Signs :

I: All positive

II: Sine positive

III: Tangent positive

IV: Cosine positive

:

ASTC

25. Important Practice Questions

1. Solve:

$$\sin x = \frac{1}{2}.$$

2. Solve:

$$\cos x = \frac{1}{2}.$$

3. Solve:

$$\tan x = 1.$$

4. Solve:

$$\sin x = 0.$$

5. Solve:

$$\cos x = 0.$$

6. Solve:

$$\tan x = 0.$$

7. Solve:

$$\sin x = -\frac{1}{2}, \quad 0 \leq x \leq 2\pi.$$

8. Solve:

$$\cos x = -\frac{\sqrt{3}}{2}, \quad 0 \leq x \leq 2\pi.$$

9. Solve:

$$\tan x = -\sqrt{3}, \quad 0 \leq x \leq 2\pi.$$

10. Solve:

$$\sin^2 x = \frac{3}{4}, \quad 0 \leq x \leq 2\pi.$$

11. Solve:

$$\cos^2 x = \frac{1}{2}, \quad 0 \leq x \leq 2\pi.$$

12. Solve:

$$\tan^2 x = 3, \quad 0 \leq x \leq 2\pi.$$

13. Find the general solution of

$$\sin x = \sin \alpha.$$

14. Find the general solution of

$$\cos x = \cos \alpha.$$

15. Find the general solution of

$$\tan x = \tan \alpha.$$

26. ONE-MINUTE REVISION

GENERAL SOLUTIONS

$$\sin x = \sin \alpha \Rightarrow x = n\pi + (-1)^n \alpha$$

$$\cos x = \cos \alpha \Rightarrow x = 2n\pi \pm \alpha$$

$$\tan x = \tan \alpha \Rightarrow x = n\pi + \alpha$$

$$\sin x = 0 \Rightarrow x = n\pi$$

$$\cos x = 0 \Rightarrow x = \frac{(2n+1)\pi}{2}$$

$$\tan x = 0 \Rightarrow x = n\pi$$

$$\sin^2 x = \sin^2 \alpha \Rightarrow x = n\pi \pm \alpha$$

$$\cos^2 x = \cos^2 \alpha \Rightarrow x = n\pi \pm \alpha$$

PART 7: IMPORTANT TRIGONOMETRIC IDENTITIES

1. What is a Trigonometric Identity?

trigonometric equation domain permissible value of the variable true , **Trigonometric Identity**

:

$$\sin^2 x + \cos^2 x = 1.$$

x permissible value true

,

$$\sin^2 x + \cos^2 x = 1$$

fundamental trigonometric identity

2. Fundamental Pythagorean Identities

important identity:

KEY FORMULA

$$\sin^2 x + \cos^2 x = 1$$

identities

sides $\cos^2 x$ divide :

$$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}.$$

Therefore,

$$\tan^2 x + 1 = \sec^2 x$$

$$1 + \tan^2 x = \sec^2 x$$

$\sin^2 x + \cos^2 x = 1$ $\sin^2 x$ divide :

$$1 + \cot^2 x = \csc^2 x$$

3. Fundamental Identity Box

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

4. Reciprocal Identities

KEY FORMULA

$$\sin x = \frac{1}{\csc x}$$

$$\cos x = \frac{1}{\sec x}$$

$$\tan x = \frac{1}{\cot x}$$

$$\csc x = \frac{1}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\cot x = \frac{1}{\tan x}$$

5. Quotient Identities

KEY FORMULA

$\tan x = \frac{\sin x}{\cos x}$

$\cot x = \frac{\cos x}{\sin x}$

formulas use identities prove

6. Fundamental Identity Network

$\sin^2 x + \cos^2 x = 1$

:

$1 - \sin^2 x = \cos^2 x$

$1 - \cos^2 x = \sin^2 x.$

:

$1 + \tan^2 x = \sec^2 x$

:

$\sec^2 x - \tan^2 x = 1.$

:

$1 + \cot^2 x = \csc^2 x$

:

$\csc^2 x - \cot^2 x = 1.$

:

IMPORTANT

$\sin^2 x = 1 - \cos^2 x$

$\cos^2 x = 1 - \sin^2 x$

$\tan^2 x = \sec^2 x - 1$

$\cot^2 x = \csc^2 x - 1$

7. Complementary Angle Identities

angles sum 90° , complementary angles
identities:

KEY FORMULA

$$\sin(90^\circ - x) = \cos x$$

$$\cos(90^\circ - x) = \sin x$$

$$\tan(90^\circ - x) = \cot x$$

$$\cot(90^\circ - x) = \tan x$$

$$\sec(90^\circ - x) = \csc x$$

$$\csc(90^\circ - x) = \sec x$$

Radians :

$$90^\circ = \frac{\pi}{2}.$$

:

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x$$

8. Even-Odd Properties

Sine odd function :

$$\sin(-x) = -\sin x$$

Cosine even function :

$$\cos(-x) = \cos x$$

Tangent:

$$\tan(-x) = -\tan x$$

:

$$\cot(-x) = -\cot x$$

$$\sec(-x) = \sec x$$

$$\csc(-x) = -\csc x$$

9. Allied Angle Identities

$$180^\circ - x$$

$$\sin(180^\circ - x) = \sin x$$

$$\cos(180^\circ - x) = -\cos x$$

$$\tan(180^\circ - x) = -\tan x$$

$$180^\circ + x$$

$$\sin(180^\circ + x) = -\sin x$$

$$\cos(180^\circ + x) = -\cos x$$

$$\tan(180^\circ + x) = \tan x$$

$$360^\circ - x$$

$$\sin(360^\circ - x) = -\sin x$$

$$\cos(360^\circ - x) = \cos x$$

$$\tan(360^\circ - x) = -\tan x$$

10. Quick Sign Table

Function	I	II	III	IV
$\sin x$	+	+	-	-
$\cos x$	+	-	-	+
$\tan x$	+	-	+	-

I: All

II: Sine

III: Tangent

IV: Cosine

11. Identity Proof: Basic Example

SOLVED EXAMPLE

Prove:

$$\frac{\sin x}{\cos x} = \tan x.$$

By definition,

$$\tan x = \frac{\sin x}{\cos x}.$$

Therefore,

$$\boxed{\frac{\sin x}{\cos x} = \tan x}$$

Hence proved.

12. Identity Proof Using Pythagorean Identity

SOLVED EXAMPLE

Prove:

$$1 + \tan^2 x = \sec^2 x.$$

We know:

$$\sin^2 x + \cos^2 x = 1.$$

Dividing by $\cos^2 x$,

$$\frac{\sin^2 x}{\cos^2 x} + 1 = \frac{1}{\cos^2 x}.$$

Therefore,

$$\tan^2 x + 1 = \sec^2 x.$$

Hence,

$$\boxed{1 + \tan^2 x = \sec^2 x}$$

13. Important Identity

Prove:

$$\frac{1 - \cos^2 x}{\sin^2 x} = 1.$$

We know:

$$1 - \cos^2 x = \sin^2 x.$$

Therefore,

$$\frac{1 - \cos^2 x}{\sin^2 x} = \frac{\sin^2 x}{\sin^2 x} = 1.$$

Hence,

$$\boxed{\frac{1 - \cos^2 x}{\sin^2 x} = 1}$$

14. Important Identity

Prove:

$$\frac{\sec x - \cos x}{\tan x} = \sin x.$$

Start with LHS:

$$\frac{\sec x - \cos x}{\tan x}.$$

Replace $\sec x$ by $\frac{1}{\cos x}$:

$$= \frac{\frac{1}{\cos x} - \cos x}{\tan x}.$$

Taking LCM,

$$= \frac{\frac{1 - \cos^2 x}{\cos x}}{\tan x}.$$

Using

$$1 - \cos^2 x = \sin^2 x,$$

we get

$$= \frac{\sin^2 x}{\cos x \tan x}.$$

Since

$$\tan x = \frac{\sin x}{\cos x},$$

we obtain

$$= \frac{\sin^2 x}{\sin x} = \sin x.$$

Therefore,

$$\boxed{\frac{\sec x - \cos x}{\tan x} = \sin x}$$

15. Important Identity

Prove:

$$\frac{\csc x - \sin x}{\cot x} = \cos x.$$

LHS:

$$\frac{\csc x - \sin x}{\cot x}.$$

Using

$$\csc x = \frac{1}{\sin x},$$

we get

$$= \frac{\frac{1}{\sin x} - \sin x}{\cot x}.$$

Taking LCM,

$$= \frac{1 - \sin^2 x}{\sin x \cot x}.$$

Since

$$1 - \sin^2 x = \cos^2 x,$$

we get

$$= \frac{\cos^2 x}{\sin x \cot x}.$$

Now,

$$\cot x = \frac{\cos x}{\sin x}.$$

Therefore,

$$= \frac{\cos^2 x}{\cos x} = \cos x.$$

Hence,

$\frac{\csc x - \sin x}{\cot x} = \cos x$

16. Factorisation Technique

identities factorisation useful technique
:

$$\sin^2 x - \cos^2 x.$$

difference of squares :

$$\sin^2 x - \cos^2 x = (\sin x - \cos x)(\sin x + \cos x).$$

:

$$\sin^2 x - \cos^2 x = (\sin x - \cos x)(\sin x + \cos x)$$

17. Important Simplification

Simplify:

$$\frac{\sin^2 x - \cos^2 x}{\sin x - \cos x}.$$

Factorise numerator:

$$\sin^2 x - \cos^2 x = (\sin x - \cos x)(\sin x + \cos x).$$

Therefore,

$$\frac{(\sin x - \cos x)(\sin x + \cos x)}{\sin x - \cos x}.$$

Hence,

$$= \sin x + \cos x$$

18. A Very Important Identity

Prove:

$$(\sin x + \cos x)^2 = 1 + \sin 2x.$$

LHS:

$$(\sin x + \cos x)^2$$

$$= \sin^2 x + \cos^2 x + 2 \sin x \cos x.$$

Using

$$\sin^2 x + \cos^2 x = 1$$

and

$$2 \sin x \cos x = \sin 2x,$$

we obtain

$$(\sin x + \cos x)^2 = 1 + \sin 2x$$

19. Another Important Identity

Similarly,

$$\begin{aligned} & (\sin x - \cos x)^2 \\ &= \sin^2 x + \cos^2 x - 2 \sin x \cos x. \end{aligned}$$

Therefore,

$$\boxed{(\sin x - \cos x)^2 = 1 - \sin 2x}$$

20. Useful Result

Adding the two identities:

$$\begin{aligned} & (\sin x + \cos x)^2 + (\sin x - \cos x)^2 \\ &= (1 + \sin 2x) + (1 - \sin 2x). \end{aligned}$$

Therefore,

$$\boxed{(\sin x + \cos x)^2 + (\sin x - \cos x)^2 = 2}$$

21. Important Identity

Prove:

$$\frac{1 + \sin x}{\cos x} = \frac{\cos x}{1 - \sin x}.$$

Start with LHS:

$$\frac{1 + \sin x}{\cos x}.$$

Multiply numerator and denominator by

$$1 - \sin x.$$

Then,

$$= \frac{(1 + \sin x)(1 - \sin x)}{\cos x(1 - \sin x)}.$$

Using

$$(1 + \sin x)(1 - \sin x) = 1 - \sin^2 x = \cos^2 x,$$

we get

$$= \frac{\cos^2 x}{\cos x(1 - \sin x)}.$$

Hence,

$$\boxed{\frac{1 + \sin x}{\cos x} = \frac{\cos x}{1 - \sin x}}$$

22. Important Identity

Prove:

$$\frac{1 + \cos x}{\sin x} = \frac{\sin x}{1 - \cos x}.$$

Rationalising the LHS:

$$\begin{aligned} \frac{1 + \cos x}{\sin x} &\cdot \frac{1 - \cos x}{1 - \cos x} \\ &= \frac{1 - \cos^2 x}{\sin x(1 - \cos x)}. \end{aligned}$$

Since

$$1 - \cos^2 x = \sin^2 x,$$

we get

$$= \frac{\sin^2 x}{\sin x(1 - \cos x)}.$$

Therefore,

$$\boxed{\frac{1 + \cos x}{\sin x} = \frac{\sin x}{1 - \cos x}}$$

23. Best Techniques for Proving Identities

IMPORTANT

Technique 1: Start from the complicated side

RHS simple, generally LHS start

Technique 2: Convert everything into sin and cos

$$\tan x = \frac{\sin x}{\cos x},$$

$$\cot x = \frac{\cos x}{\sin x},$$

$$\sec x = \frac{1}{\cos x},$$

$$\csc x = \frac{1}{\sin x}.$$

Technique 3: Use fundamental identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x.$$

Technique 4: Factorise

$$a^2 - b^2 = (a - b)(a + b).$$

Technique 5: Rationalise expression

$$1 + \sin x, \quad 1 - \sin x$$

$$1 + \cos x, \quad 1 - \cos x$$

, conjugate multiply useful

24. Common Mistakes

1. Identity prove LHS RHS manipulate

2. $\sin^2 x \neq \sin(x^2)$

3.

$$\sin(A + B) \neq \sin A + \sin B.$$

4.

$$\cos(A + B) \neq \cos A + \cos B.$$

5. Denominator zero possibility ignore
6. Square root sign

25. Important Practice Questions

Prove that $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$.

Alternate

Find the value of $\tan \frac{\pi}{2}$.

Prove that $\cos 6x = 32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1$.

Given $\tan x = \frac{2}{7}$ and x lies in the third quadrant. Based on this information, answer the following questions:

1. Find the value of $\sin 2x$.
2. Find the value of $\cos 2x$.
3. Find the value of $\sin \frac{x}{2}$.

OR

Find the value of $\cos \frac{x}{2}$.

26. ONE-MINUTE REVISION

MUST REMEMBER IDENTITIES

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\cot x = \frac{\cos x}{\sin x}$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = 1 - 2 \sin^2 x$$

$$\cos 2x = 2 \cos^2 x - 1$$

$$(\sin x + \cos x)^2 = 1 + \sin 2x$$

$$(\sin x - \cos x)^2 = 1 - \sin 2x$$

CLASS 11 TRIGONOMETRY

ONE-PAGE COMPLETE MIND MAP

1. BASIC CONCEPTS

$$180^\circ = \pi \text{ rad}$$

$$360^\circ = 2\pi \text{ rad}$$

$$\theta^\circ = \frac{\theta\pi}{180} \text{ rad}$$

$$\theta \text{ rad} = \frac{180\theta^\circ}{\pi}$$

Six Functions:

$$\sin x, \cos x, \tan x$$

$$\csc x, \sec x, \cot x$$

2. TRIGONOMETRIC RATIOS

$$\sin x = \frac{P}{H}$$

$$\cos x = \frac{B}{H}$$

$$\tan x = \frac{P}{B}$$

$$\csc x = \frac{1}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\cot x = \frac{1}{\tan x}$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\cot x = \frac{\cos x}{\sin x}$$

3. FUNDAMENTAL IDENTITIES

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\cos^2 x = 1 - \sin^2 x$$

$$\tan^2 x = \sec^2 x - 1$$

$$\cot^2 x = \csc^2 x - 1$$

4. QUADRANT SIGNS

ASTC

I: All +

II: Only $\sin +$

III: Only $\tan +$

IV: Only $\cos +$

5. STANDARD VALUES

x	0°	30°	45°	60°	90°
$\sin x$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND

6. ALLIED ANGLES

$$\sin(90^\circ - x) = \cos x$$

$$\cos(90^\circ - x) = \sin x$$

$$\tan(90^\circ - x) = \cot x$$

$$\sin(180^\circ - x) = \sin x$$

$$\cos(180^\circ - x) = -\cos x$$

$$\tan(180^\circ - x) = -\tan x$$

$$\sin(360^\circ - x) = -\sin x$$

$$\cos(360^\circ - x) = \cos x$$

$$\tan(360^\circ - x) = -\tan x$$

7. COMPOUND ANGLES

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

8. MULTIPLE ANGLES

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = 1 - 2 \sin^2 A$$

$$\cos 2A = 2 \cos^2 A - 1$$

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

9. HALF ANGLE

$$\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$$

$$\cos^2 \frac{A}{2} = \frac{1 + \cos A}{2}$$

$$\tan \frac{A}{2} = \frac{\sin A}{1 + \cos A}$$

$$\tan \frac{A}{2} = \frac{1 - \cos A}{\sin A}$$

10. TRANSFORMATION FORMULAE

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

11. PRODUCT $\rightarrow$ SUM

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

12. TRIGONOMETRIC EQUATIONS

$$\sin x = \sin \alpha \Rightarrow x = n\pi + (-1)^n \alpha$$

$$\cos x = \cos \alpha \Rightarrow x = 2n\pi \pm \alpha$$

$$\tan x = \tan \alpha \Rightarrow x = n\pi + \alpha$$

$$n \in \mathbb{Z}$$

13. PERIODICITY

$$\sin(x+2\pi) = \sin x$$

$$\cos(x+2\pi) = \cos x$$

$$\tan(x+\pi) = \tan x$$

$$T_{\sin x} = 2\pi, \quad T_{\cos x} = 2\pi, \quad T_{\tan x} = \pi$$

14. IDENTITY PROOF TRICKS

1. Complicated side
2. $\tan, \cot, \sec, \csc$ $\sin, \cos$
3. Fundamental identities
4. Factorisation
5. LCM
6. Conjugate rationalise

$$\text{LHS} \rightarrow \text{RHS}$$

MASTER FORMULA BOX

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = 1 - 2 \sin^2 A$$

$$\cos 2A = 2 \cos^2 A - 1$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

TRIGONOMETRY = RATIOS + IDENTITIES + FORMULAE + EQUATIONS